

Introduction to PowerBI & Power Query

Hands-On Tutorial: Northwind Sales Analysis via OData

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Project 03 – Multiple Source Data

M.Sc. in Electrical and Computer Engineering – ISE – Universidade do Algarve

Data source: Northwind OData feed –

<https://services.odata.org/V3/Northwind/Northwind.svc/>

Business Scenario

You are a data analyst at Northwind Traders, a fictitious food-and-beverage distributor. Management has asked for a unified sales dashboard that answers key operational questions:

- Which product categories drive the most revenue?
- Which customers and ship-to countries generate the highest order value?
- How do individual employees perform in terms of sales?
- How do sales evolve over time?

All data is available through the company's OData service. The final report should combine **8 tables** loaded from a single OData endpoint into a clean star-schema model and deliver at least four interactive visuals.

Learning Outcomes

By the end of this tutorial, you will be able to:

- Connect to an OData feed and load multiple tables in one operation.
- Remove navigation-property columns (type Table/Record) that OData adds automatically.
- Create calculated columns in Power Query (LineTotal, FullName).
- Verify and, if necessary, manually configure a multi-table star-schema model.
- Build interactive visuals that span several tables via relationships.
- Interpret cross-filter behaviour and formulate data-driven recommendations.

Data Source

- **Protocol:** OData v3
- **URL:** <https://services.odata.org/V3/Northwind/Northwind.svc/>
- **Tables to load** (8 total):

Table	Role	Key column(s)
Orders	Fact (header)	OrderID
Order_Details	Fact (lines)	OrderID, ProductID
Products	Dimension	ProductID
Categories	Dimension	CategoryID
Suppliers	Dimension	SupplierID
Customers	Dimension	CustomerID
Employees	Dimension	EmployeeID
Shippers	Dimension	ShipperID

Prerequisites

- Power BI Desktop installed.
- Internet access to reach services.odata.org.
- No local files required – everything comes from the OData feed.
- Basic familiarity with Power Query (Applied Steps, column operations).

Lab Workflow Overview

1. Connect to the OData feed and select all 8 tables.
2. Clean each query: remove navigation-property columns and keep only the relevant fields.
3. Add computed columns: LineTotal in Order_Details; FullName in Employees.
4. Close & Apply; verify auto-detected relationships and add any missing ones.
5. Build four report visuals and interact with cross-filtering.
6. Create KPI measures and add card visuals.
7. Answer the analytical questions and save/publish the report.

Step 1 – Connect to the OData Feed

1. Open **Power BI Desktop**.
2. **Home** → **Get data** → **OData feed**.
3. Enter the URL:

<https://services.odata.org/V3/Northwind/Northwind.svc/>

4. Click **OK**. In the Navigator, tick the following tables:

Categories, Customers, Employees, Order_Details, Orders, Products, Shippers, Suppliers

5. Click **Transform Data** to open Power Query Editor.

Note on navigation properties

OData automatically includes navigation-property columns (shown as Table or Record in the preview). These are hyperlinks to related entities – they are *not* the foreign-key integer values you need. You will remove them in the cleaning steps below.

Step 2 – Clean the Fact Tables

2a – Order_Details

Select the Order_Details query in the left panel.

1. **Remove** all columns of type Table or Record (right-click header → **Remove**).
2. Keep the following columns:
 - OrderID (Whole Number)
 - ProductID (Whole Number)
 - UnitPrice (Fixed decimal number)
 - Quantity (Whole Number)
 - Discount (Decimal Number)
3. Use **Remove Other Columns** to drop everything else.
4. Verify data types match the list above; correct any mismatches via the column-type icon.

Add Computed Column: LineTotal

1. **Add Column** → **Custom Column**.
2. Name: LineTotal.
3. Formula:

$$[\text{UnitPrice}] * [\text{Quantity}] * (1 - [\text{Discount}])$$

4. Click **OK**.
5. Set the data type of LineTotal to **Fixed decimal number**.

2b – Orders

Select the Orders query.

1. Remove all Table/Record navigation columns.
2. Keep:
 - OrderID (Whole Number)
 - CustomerID (Text)
 - EmployeeID (Whole Number)
 - OrderDate (Date)
 - RequiredDate (Date)
 - ShippedDate (Date)
 - ShipVia (Whole Number) – links to Shippers[ShipperID]
 - Freight (Fixed decimal number)
 - ShipCity (Text)
 - ShipCountry (Text)
3. Use **Remove Other Columns**.
4. Confirm data types.

Checkpoint

Order_Details should have 6 columns (5 original + LineTotal). Orders should have 10 columns. Check the *Applied Steps* panel on the right – each action you performed should be listed there.

Step 3 – Clean the Dimension Tables

For each dimension table below, remove all Table/Record columns and keep only the listed fields. Use **Remove Other Columns** after selecting the columns to keep.

Products

Keep:

- ProductID, ProductName, CategoryID, SupplierID, QuantityPerUnit, UnitPrice, UnitsInStock, Discontinued

Set UnitPrice → Fixed decimal; UnitsInStock → Whole Number; Discontinued → True/False (Logical).

Categories

Keep: CategoryID, CategoryName, Description

Suppliers

Keep: SupplierID, CompanyName, ContactName, City, Country

Customers

Keep: CustomerID, CompanyName, ContactName, City, Country

Employees

1. Keep: EmployeeID, FirstName, LastName, Title, City, Country
2. Remove all other columns (including navigation columns such as Employees1, Employees2, Orders).
3. **Add a computed column** FullName:
 - **Add Column** → **Custom Column**.
 - Name: FullName.
 - Formula: [FirstName] & " " & [LastName]

Shippers

Keep: ShipperID, CompanyName, Phone

Checkpoint

After cleaning, the approximate column counts should be: Products (8), Categories (3), Suppliers (5), Customers (5), Employees (7, including FullName), Shippers (3). Scroll through each table preview and confirm the data looks sensible.

Step 4 – Close & Apply and Verify the Model

1. In Power Query Editor, click **Home** → **Close & Apply**.
2. After loading completes, switch to the **Model view** (diagram icon on the left rail).
3. Power BI should auto-detect most relationships. Verify the following are active (single arrow = many-to-one direction):

Many side	One side	Join key
Order_Details[OrderID]	Orders[OrderID]	OrderID
Order_Details[ProductID]	Products[ProductID]	ProductID
Orders[CustomerID]	Customers[CustomerID]	CustomerID
Orders[EmployeeID]	Employees[EmployeeID]	EmployeeID
Orders[ShipVia]	Shippers[ShipperID]	ShipperID
Products[CategoryID]	Categories[CategoryID]	CategoryID
Products[SupplierID]	Suppliers[SupplierID]	SupplierID

4. For any missing relationship: click **Manage relationships** → **New** and configure the key columns manually.
5. Confirm cardinality is **Many to One** (*:1) and cross-filter direction is **Single** for all relationships.
6. Figure 1 shows an example of the final data model in Model view.

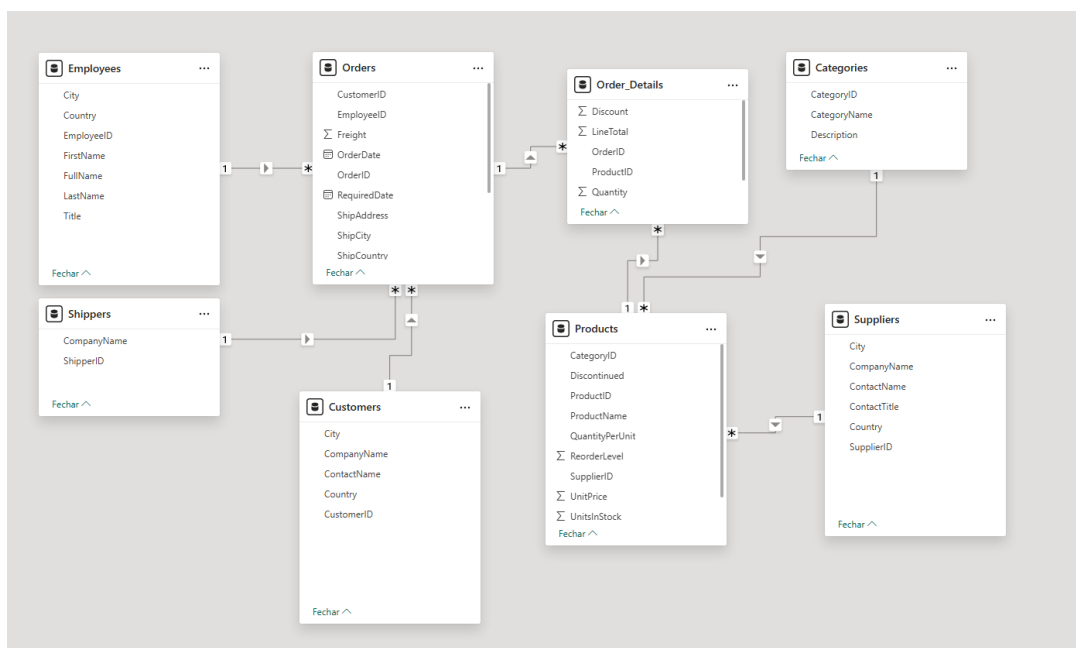


Figure 1: Example of the final data model in Model view.

Step 5 – Build the Report Visuals

Switch to **Report view**. Build the four visuals below on the same report page (or on separate pages – your choice).

Visual 1: Revenue by Product Category (Bar Chart)

1. Insert a **Clustered bar chart**.
2. **Y-axis:** Categories[CategoryName].
3. **X-axis:** Order_Details[LineTotal] (set aggregation to **Sum**).
4. Sort bars descending by LineTotal.
5. Title: *Revenue by Category*.

Visual 2: Revenue over Time (Line Chart)

1. Insert a **Line chart**.
2. **X-axis:** Orders[OrderDate] – in the *Visualizations* pane, expand the date hierarchy down to **Month** level for a cleaner view.
3. **Y-axis:** Order_Details[LineTotal] (Sum).
4. Title: *Monthly Revenue Trend*.

Visual 3: Revenue by Ship-to Country (Map)

1. Insert a **Map** visual (or **Filled map**).
2. **Location:** Orders[ShipCountry].

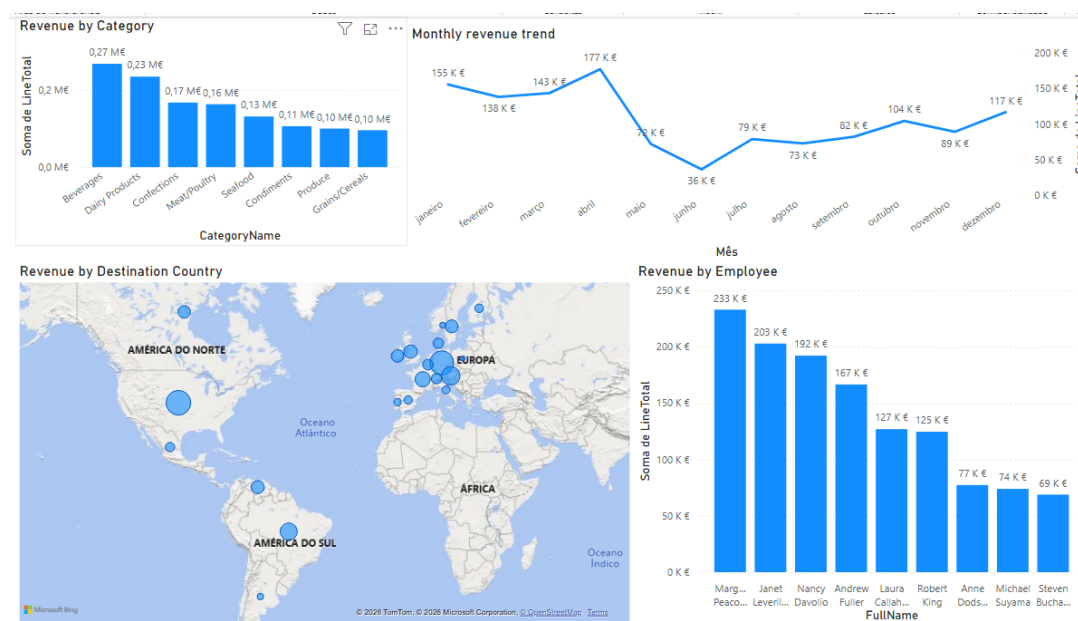


Figure 2: Example of the final report view with all four visuals.

3. **Bubble size** (or **Color saturation**): `Order_Details[LineTotal]` (Sum).
4. Title: *Revenue by Destination Country*.

Visual 4: Employee Sales Performance (Bar Chart)

1. Insert a **Clustered column chart**.
2. **X-axis**: `Employees[FullName]`.
3. **Y-axis**: `Order_Details[LineTotal]` (Sum).
4. Sort descending.
5. Title: *Revenue by Employee*.

Step 6 – Interact and Analyse

1. Click on a bar in *Visual 1* (e.g., *Beverages*) and observe how the other visuals cross-filter.
2. Click on a country in *Visual 3* and check which employees and categories dominate that market.
3. Click on an employee in *Visual 4* and inspect which categories and months show the highest activity.
4. Hold **Ctrl** and click multiple items to build compound filter selections.
5. Press **Esc** or click on blank canvas to clear all selections.

Figure 2 shows the final report view with all four visuals.

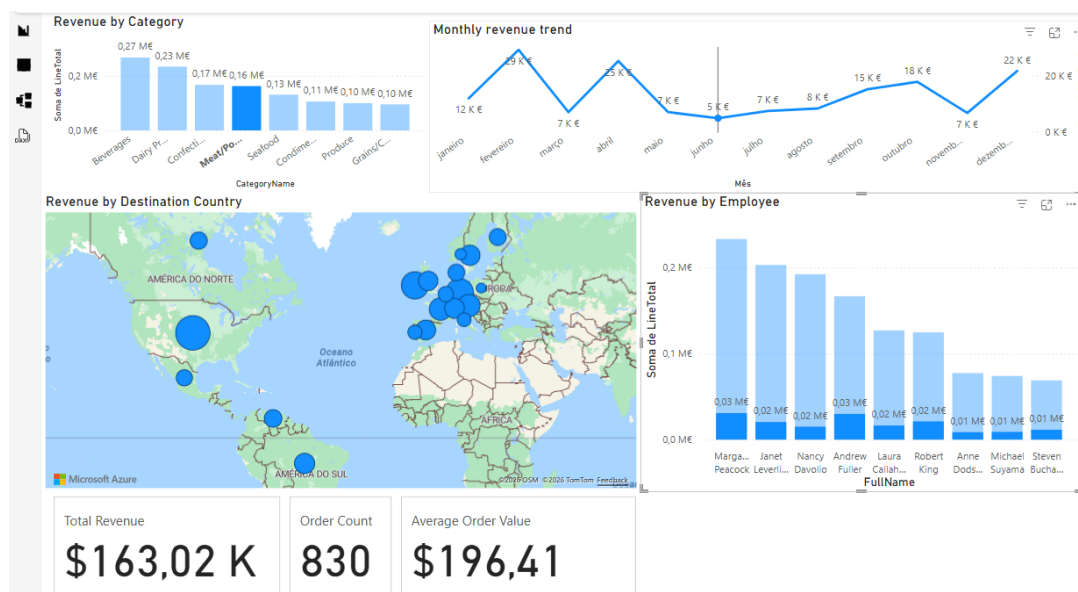


Figure 3: Example of the final report view with all four visuals.

Step 7 – Create KPI Measures and Cards

Create measures in Order_Details and Orders.

Total Revenue

```
Total Revenue = SUM(Order_Details[LineTotal])
```

Order Count

```
Order Count = DISTINCTCOUNT(Orders[OrderID])
```

Average Order Value

```
Average Order Value = DIVIDE([Total Revenue], [Order Count], 0)
```

Then add **Card** visuals to the report canvas and set these measures as their values.

Figure 3 shows the final report view with all four visuals including the new KPI cards.

Format: currency for revenue measures, whole number for counts, percentage for ratio measures.

Extra – Advanced DAX Measures (Optional)

Use this section after Step 7 if you want to extend the report with context-aware and time-intelligence KPIs.

Extra 1: Context-Based Measures

Revenue (Beverages)

```
Revenue Beverages = CALCULATE([Total Revenue], Categories[CategoryName] = "Beverages")
```


Revenue (USA)

```
Revenue USA = CALCULATE( [Total Revenue], Orders[ShipCountry]
= "USA" )
```

Revenue Share % (Category vs Total)

```
Revenue Share % = DIVIDE( [Total Revenue], CALCULATE([Total
Revenue], ALL(Categories)), 0 )
```

Top Employee Revenue

```
Top Employee Revenue = MAXX( VALUES(Employees[FullName]),
[Total Revenue] )
```

Questions to Answer with the Report

Answer each question in 1–3 sentences. Cite which visual(s) you used as evidence.

1. Which product category generates the highest total revenue? Is the difference between the top and bottom categories significant?
2. Which three countries receive the largest order values? How does the map visual help communicate this finding compared to a table?
3. Identify a period of notably high or low monthly revenue. Suggest a possible business explanation.
4. Which two employees have the highest sales totals? Does cross-filtering by category reveal any specialisation in their order mix?
5. Pick one supplier (Suppliers table). Use the model relationships to determine how many distinct products that supplier provides and in which category most of them fall. (Hint: use a Table visual with slicers.)
6. If you were advising Northwind on where to focus marketing spend next quarter, what market (country/category combination) would you recommend and why?

Step 8 – Save and Publish

1. Save your report: Ctrl + S → Northwind_Sales.pbix.
2. **Home** → **Publish**.
3. Select your destination workspace in Power BI Service.
4. Open the published report in the browser and verify the visuals render correctly.

A Appendix – DAX Function Reference for Extra Measures

Syntax (generic)	What it does
<code>CALCULATE(expression, filter1, ...)</code>	Evaluates an expression under a modified filter context.
<code>ALL(table_or_column)</code>	Removes filters from a table/column to compute grand totals or baseline values.
<code>DIVIDE(numerator, denominator, alternateResult)</code>	Safe division with fallback for zero/blank denominator.
<code>MAXX(table, expression)</code>	Iterates rows of a table expression and returns the maximum evaluated result.
<code>VALUES(column)</code>	Returns the distinct values visible in current filter context.
<code>CALENDAR(start_date, end_date)</code>	Creates a contiguous date table between two dates.
<code>MIN(column)</code>	Returns the smallest value in a column.
<code>MAX(column)</code>	Returns the largest value in a column.
<code>TOTALYTD(expression, dates)</code>	Aggregates expression from start of year to current context date.
<code>SAMEPERIODLASTYEAR(dates)</code>	Returns the equivalent date range shifted one year back.

For reliable time intelligence: use a dedicated date table, mark it as Date table in the model, and ensure a valid relationship to `Orders[OrderDate]`.

Troubleshooting Notes

- **OData load fails:** verify internet access and that the URL contains no trailing spaces. Try pasting the URL into a browser – you should see an XML/JSON service document.
- **Navigation-property columns still present:** in Power Query, identify columns of type `Table` or `Record` and remove them, then re-run *Remove Other Columns*.
- **Relationship not auto-detected:** open *Manage relationships*, click *Autodetect*, then manually add any still missing using the table in Step 4.
- **Map visual shows no data:** ensure `Orders[ShipCountry]` is categorised as *Country* in **Column tools** → **Data category**.
- **Date hierarchy groups unexpectedly:** in the line chart, select `OrderDate` without hierarchy, or drill down to the *Month* level manually.
- **LineTotal shows zero or blank:** confirm that `Discount` is typed as *Decimal Number* (not *Text*) and re-evaluate the custom column formula.

Key Takeaways

- A single OData endpoint can provide an entire normalised database; no local files are needed.
- Navigation-property columns must be removed before loading to avoid type errors and model bloat.

- A star schema (fact tables + dimension tables connected by keys) is the foundation of efficient Power BI models.
- Computed columns in Power Query (`LineTotal`, `FullName`) keep business logic centralised and reusable.
- Cross-filter propagation through relationships turns independent tables into a coherent, explorable dataset – enabling insights that no single table could provide alone.

End of tutorial – Key Takeaways completed